



*Open-Source SIEM & XDR Security Platform*

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Presented to the College of Information Technology and Engineering

Information Technology Department

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PUBLIC REPOSITORY COPY ? Credentials and default passwords have been redacted.

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## Executive Summary

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This document presents Wazuh, a free, open-source unified security platform, as the chosen tool for the IT Elective 3 Ethical Hacking final project. Wazuh provides Security Information and Event Management (SIEM), Endpoint Detection and Response (EDR), and Extended Detection and Response (XDR) capabilities in a single, cohesive solution, making it one of the most comprehensive open-source security platforms available today.

For this project, Wazuh v4.10.1 was successfully deployed using Docker in a single-node configuration on a *Debian Linux* virtual machine hosted in the cloud. All four core components were configured and are fully operational: the Wazuh Manager (analysis engine), Wazuh Indexer (data storage), Wazuh Dashboard (web interface), and Wazuh Agents (endpoint collectors). The live environment currently monitors active agents and generated medium-severity alerts in the last 24 hours at the time of presentation.

This documentation covers the complete lifecycle of the tool, from installation and deployment to live demonstrations of core capabilities including Vulnerability Detection, File Integrity Monitoring, and Security Configuration Assessment. The demonstrations are designed to highlight the relevance of SIEM/XDR tooling in an ethical hacking context by showing both the attack and the detection perspective.

| Tool Name       | Wazuh                           |
|-----------------|---------------------------------|
| Version         | 4.10.1                          |
| Deployment Type | Docker Single-Node              |
| Host OS         | Debian Linux (Cloud VM)         |
| Active Agents   | 4                               |
| Alerts (24h)    | 74 Medium / 0 High / 0 Critical |
| Framework       | MITRE ATT&CK                    |



# Introduction to Wazuh

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## Background

The cybersecurity landscape demands tools capable of detecting, analyzing, and responding to threats in real time across complex, distributed environments. Traditional security approaches, siloed firewalls, isolated antivirus, and manual log review, are no longer sufficient. Modern security operations require platforms that can correlate data from multiple sources, apply intelligent rules, and provide actionable insights.

Wazuh was originally forked from OSSEC (Open Source HIDS Security) in 2015 and has since evolved far beyond its origins as a host-based intrusion detection system. It is now a full-featured, enterprise-grade security platform trusted by thousands of organizations globally, including Fortune 500 companies and government agencies. Its open-source nature, combined with enterprise capabilities, makes it an ideal subject for academic study in ethical hacking.

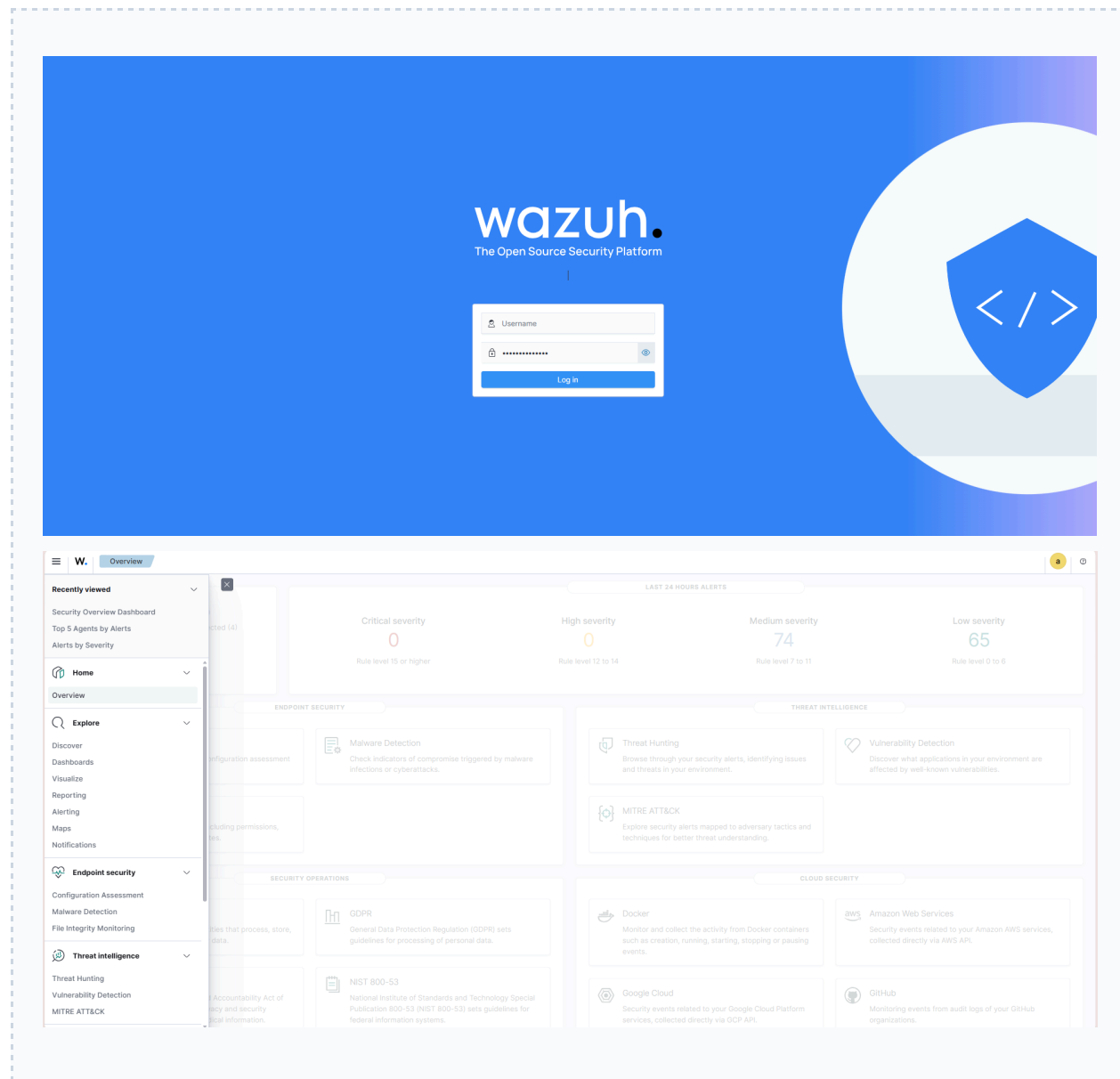
## What is Wazuh?

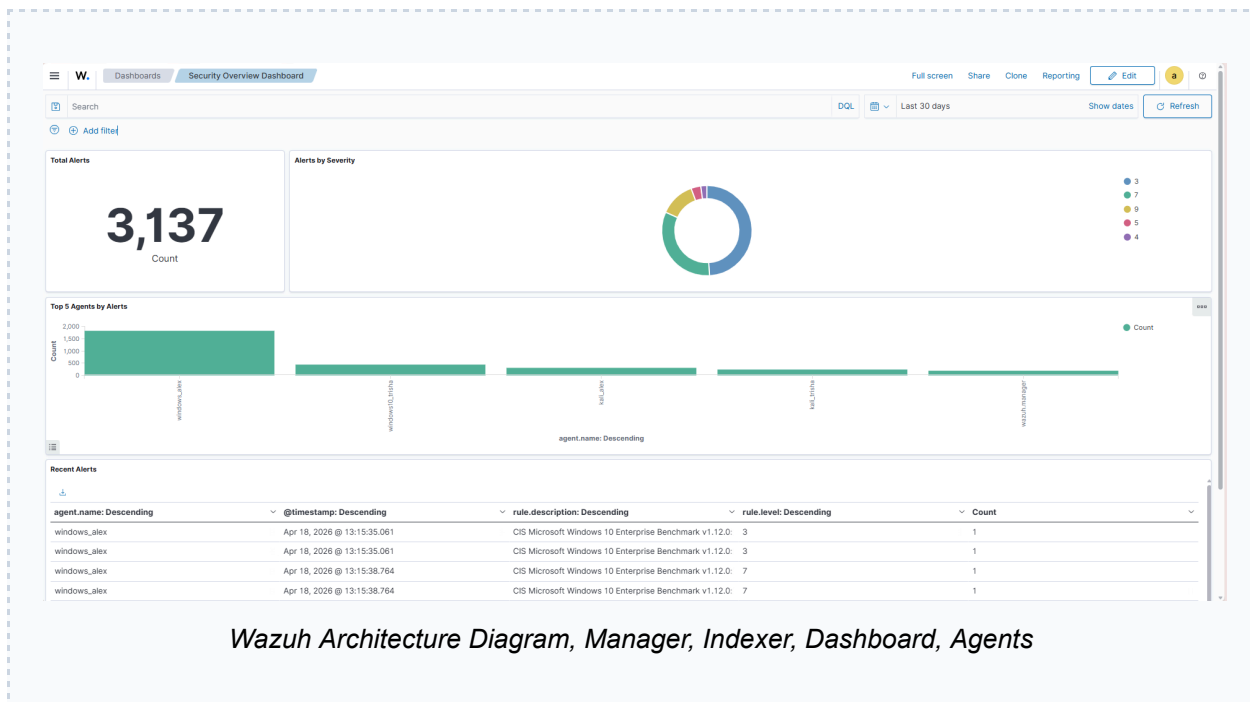
Wazuh is a free, open-source unified cybersecurity platform that delivers XDR and SIEM capabilities. It protects workloads across on-premises, virtualized, containerized, and cloud-based environments. From a single dashboard, it provides real-time threat detection, log analysis, vulnerability scanning, compliance monitoring, and automated incident response. It bridges the gap between offensive tactics and defensive visibility. By providing real-time threat monitoring, the platform allows users to see exactly how attacks are logged and alerted from a defender's perspective. This insight is deepened through automatic mapping to the MITRE ATT&CK framework, the global standard for threat intelligence. Understanding such a toolset enables aspiring penetration testers to craft stealthier attacks and deliver more actionable findings to clients. Furthermore, Wazuh exposes users to the regulatory side of cybersecurity through built-in compliance modules for PCI DSS, GDPR, HIPAA, and NIST 800-53. As a free and open-source solution, it provides unrestricted access to source code and configurations, making it an ideal environment for deep, hands-on learning without the barrier of licensing costs.

# Tool Architecture & Components

## Architecture Overview

Wazuh follows a centralized, layered architecture where lightweight agents deployed on endpoints forward security data to a central server stack for processing, storage, and visualization. All inter-component communication is secured with TLS/SSL certificates generated during deployment.





## Core Components

### Wazuh Manager

The Wazuh Manager is the analysis and orchestration engine at the heart of the platform. It receives security data from connected agents, applies a rule-based detection engine with over 3,000 built-in rules, and generates alerts when suspicious events are identified. The Manager also handles agent enrollment, configuration distribution, and Active Response actions.

### Wazuh Indexer

Built on OpenSearch (the open-source fork of Elasticsearch), the Wazuh Indexer is a high-performance, distributed search and analytics engine. It stores all normalized events, alerts, and metadata generated by the Manager and serves as the data backend for all Dashboard queries and visualizations. It provides near-real-time data availability and supports large-scale deployments.

### Wazuh Dashboard

The Wazuh Dashboard is the web-based graphical control center built on OpenSearch Dashboards. It provides security analysts with real-time visualizations, alert management, compliance reports, agent configuration, and threat intelligence views. The Dashboard is the primary interface for day-to-day security operations and the primary interface used in this project's demonstrations.

## Wazuh Agent

Wazuh Agents are lightweight, multi-platform programs (C-based, minimal footprint) installed on monitored endpoints. Agents perform local data collection , system logs, file changes, process activity, network connections, registry changes (Windows) , and forward normalized data to the Manager over an encrypted channel on TCP port 1514.

# System Requirements

---

## Hardware Requirements (Single-Node Docker)

| Resource  | Minimum         | Recommended (This Project)                  |
|-----------|-----------------|---------------------------------------------|
| CPU Cores | 2 cores         | 4 cores (Cloud VM)                          |
| RAM       | 4 GB            | 8 GB or higher                              |
| Storage   | 50 GB           | 100 GB SSD                                  |
| Network   | Stable internet | Low-latency cloud network                   |
| OS        | Debian 10+      | Debian 12 (Bookworm) , Used in this project |

## Software Requirements

| Software       | Version  | Purpose                                                 |
|----------------|----------|---------------------------------------------------------|
| Docker Engine  | 24.0+    | Container runtime for all Wazuh components              |
| Docker Compose | v2.0+    | Multi-container orchestration and lifecycle management  |
| Git            | Latest   | Cloning the official Wazuh Docker repository            |
| Debian Linux   | 11 or 12 | Host operating system (CLI only in this project)        |
| curl / wget    | Latest   | Downloading packages and verification scripts           |
| OpenSSL        | Latest   | Generating TLS certificates for component communication |

## Required Network Ports

| Port  | Protocol | Service                              | Direction         |
|-------|----------|--------------------------------------|-------------------|
| 1514  | TCP/UDP  | Agent to Manager communication       | Inbound (Manager) |
| 1515  | TCP      | Agent enrollment / auto-registration | Inbound (Manager) |
| 55000 | TCP      | Wazuh REST API                       | Inbound/Outbound  |
| 9200  | TCP      | Wazuh Indexer (OpenSearch) API       | Internal only     |
| 9300  | TCP      | Indexer cluster node communication   | Internal only     |
| 443   | TCP      | Wazuh Dashboard (HTTPS)              | Inbound (Users)   |

## Installation & Deployment

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### Deployment Overview

Wazuh v4.10.1 was deployed using the official Docker single-node configuration. This approach packages the Manager, Indexer, and Dashboard as a unified Docker Compose stack, reducing installation complexity while maintaining full platform functionality. The deployment runs on a Debian Linux virtual machine hosted on a cloud provider, accessible via SSH from any location.

### Step-by-Step Installation Guide

#### Step 1: Update the System

Before installing any packages, update the Debian system to ensure all base packages are current and security patches are applied:

```
$ sudo apt-get update && sudo apt-get upgrade -y
```

#### Step 2: Install Docker Engine & Docker Compose

Add the official Docker repository and install Docker Engine along with the Docker Compose plugin:

```
$ sudo apt-get install -y ca-certificates curl gnupg
$ sudo install -m 0755 -d /etc/apt/keyrings
$ curl -fsSL https://download.docker.com/linux/debian/gpg | sudo gpg --dearmor \
  -o /etc/apt/keyrings/docker.gpg
$ echo "deb [arch=$(dpkg --print-architecture) \
  signed-by=/etc/apt/keyrings/docker.gpg] \
  https://download.docker.com/linux/debian \
  $(. /etc/os-release && echo $VERSION_CODENAME) stable" | \
  sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
```

```
$ sudo apt-get update
$ sudo apt-get install -y docker-ce docker-ce-cli containerd.io
docker-compose-plugin
```

### Step 3: Increase Virtual Memory Limit

The Wazuh Indexer (OpenSearch) requires a higher virtual memory map count than the Linux default. This must be set persistently:

```
# Apply immediately (temporary):
$ sudo sysctl -w vm.max_map_count=262144

# Apply permanently (survives reboot):
$ echo "vm.max_map_count=262144" | sudo tee -a /etc/sysctl.conf
```

### Step 4: Clone the Wazuh Docker Repository

Clone the official Wazuh Docker repository and navigate into the single-node deployment directory:

```
$ git clone https://github.com/wazuh/wazuh-docker.git -b v4.10.1
$ cd wazuh-docker/single-node
```

### Step 5: Generate TLS/SSL Certificates

Generate self-signed certificates for secure internal communication between the Manager, Indexer, and Dashboard:

```
$ docker compose -f generate-indexer-certs.yml run --rm generator

# Certificates will be written to: ./config/wazuh_indexer_ssl_certs/
```

### Step 6: Deploy the Full Wazuh Stack

Start all Wazuh services in detached mode (background) using Docker Compose. Docker will automatically pull all required images on first run:

```
$ docker compose up -d

# Expected pull/start sequence:
#   Pulling wazuh/wazuh-manager:4.10.1    ... done
#   Pulling wazuh/wazuh-indexer:4.10.1    ... done
#   Pulling wazuh/wazuh-dashboard:4.10.1  ... done
```

```
# Creating single-node-wazuh.manager-1 ... done
# Creating single-node-wazuh.indexer-1 ... done
# Creating single-node-wazuh.dashboard-1 ... done
```

## Step 7: Verify Container Status

Confirm all three containers are running and healthy:

```
$ docker compose ps
```

| NAME                          | IMAGE                        | STATUS       |
|-------------------------------|------------------------------|--------------|
| single-node-wazuh.manager-1   | wazuh/wazuh-manager:4.10.1   | Up (healthy) |
| single-node-wazuh.indexer-1   | wazuh/wazuh-indexer:4.10.1   | Up (healthy) |
| single-node-wazuh.dashboard-1 | wazuh/wazuh-dashboard:4.10.1 | Up (healthy) |

## Step 8: Access the Dashboard

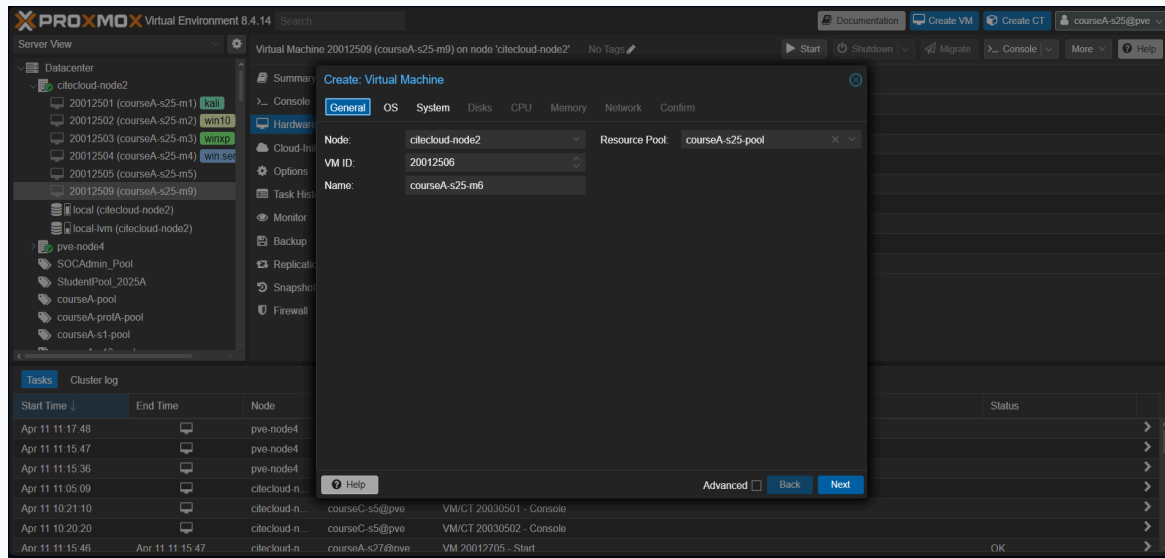
Open a web browser and navigate to the cloud VM's public IP address. Accept the self-signed certificate warning and log in with the default credentials:

```
URL:      https://<cloud-vm-public-ip>
Username:  admin
Password:  
```

```
# Note: Change the default password immediately after first login.
```

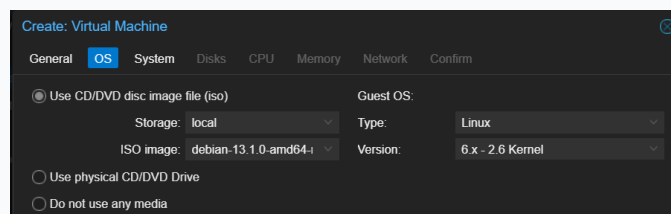
## INSTALLATION OF WAZUH

### PROXMOX

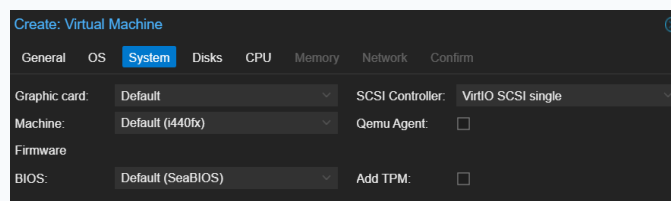


Upon accessing the Proxmox web interface, click `citectcloud-node2` and then the button of Create VM located at the top left. After this, a pop will initialize and then you will be assigning a unique VM ID, and designating a specific name to ensure it is easily identifiable.

For the ISO image, we will choose `debian 13.1.0-amd64-netinst.iso`



No changes will be made for the System





After that we will proceed to Disks, where we will put 50 GB as our disk size

The screenshot shows the 'Create: Virtual Machine' window with the 'Disks' tab selected. The window has tabs for General, OS, System, Disks, CPU, Memory, Network, and Confirm. On the left, there is a list of disks with 'scsi0' selected. Below the list is an 'Add' button. The main area shows the configuration for 'scsi0' with the following settings:

| Setting         | Value                               |
|-----------------|-------------------------------------|
| Bus/Device      | SCSI 0                              |
| Cache           | Default (No cache)                  |
| SCSI Controller | VirtIO SCSI single                  |
| Storage         | local-lvm                           |
| Disk size (GiB) | 50                                  |
| Format          | Raw disk image (raw)                |
| Discard         | <input type="checkbox"/>            |
| IO thread       | <input checked="" type="checkbox"/> |

At the bottom right, there are buttons for 'Advanced' (unchecked), 'Back', and 'Next'.

After that we will proceed to CPU, where we will put 1 - sockets and 2 - cores

The screenshot shows the 'Create: Virtual Machine' window with the 'CPU' tab selected. The window has tabs for General, OS, System, Disks, CPU, Memory, Network, and Confirm. The main area shows the configuration for the CPU with the following settings:

| Setting     | Value         |
|-------------|---------------|
| Sockets     | 1             |
| Type        | x86-64-v2-AES |
| Cores       | 2             |
| Total cores | 2             |

For our Memory, we will type it into 6144

Create: Virtual Machine

General OS System Disks CPU **Memory** Network Confirm

Memory (MiB): 6144

The rest of the process will remain unchanged. Once you reach the Confirm tab and verify the configuration or finalize the process; your new virtual machine will then be initialized and visible in the resource tree.

Create: Virtual Machine

General OS System Disks CPU Memory **Network** Confirm

☐ No network device

Bridge: vmbr2001 Model: VirtIO (paravirtualized)

VLAN Tag: no VLAN MAC address: auto

Firewall: ☒

Create: Virtual Machine

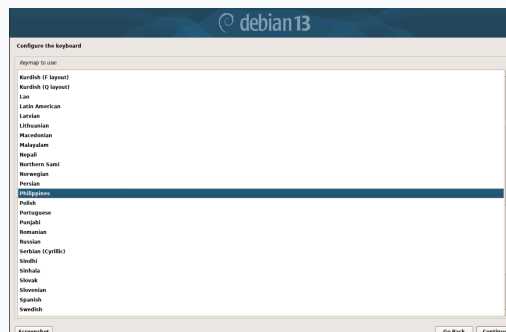
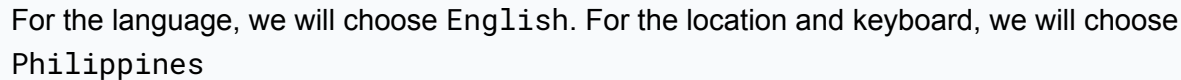
General OS System Disks CPU Memory Network **Confirm**

| Key ↑    | Value                                                 |
|----------|-------------------------------------------------------|
| cores    | 2                                                     |
| cpu      | x86-64-v2-AES                                         |
| ide2     | local:iso/debian-13.1.0-amd64-netinst.iso,media=cdrom |
| memory   | 6144                                                  |
| name     | courseA-s25-m6                                        |
| net0     | virtio,bridge=vmbr2001,firewall=1                     |
| nodename | citecloud-node2                                       |
| numa     | 0                                                     |
| ostype   | l26                                                   |
| pool     | courseA-s25-pool                                      |
| scsi0    | local-lvm:50,ioread=on                                |
| scsihw   | virtio-scsi-single                                    |
| sockets  | 1                                                     |
| vmid     | 20012506                                              |

☐ Start after created

Advanced ☐ Back Finish

After launching the new virtual machine, we will install Debian13 through Graphic Install





Next, configure the system's identity and administrative credentials. Set the Host Name to Wazuh and the Domain Name to school.local. And then, use the password as [REDACTED FOR PUBLIC REPOSITORY] then use the display name Wazuh ITELECT3



Configure the network

Please enter the hostname for this system.

The hostname is a single word that identifies your system to the network. If you don't know what your hostname should be, consult your network administrator. If you are setting up your own home network, you can make something up here.

Hostname:

wazuh



Configure the network

The domain name is the part of your Internet address to the right of your host name. It is often something that ends in .com, .net, .edu, or .org. If you are setting up a home network, you can make something up, but make sure you use the same domain name on all your computers.

Domain name:

school.local



Set up users and passwords

Some account needs to be available with administrative super-user privileges. The password for that account should be something that cannot be guessed.

To allow direct password-based access via the 'root' account, you can set the password for that account here.

Alternatively, you can lock the root account's password by leaving this setting empty, and instead use the system's initial user account (which will be set up in the next step) to gain administrative privileges. This will be enabled for you by adding that initial user to the 'sudo' group.

Note: what you type here will be hidden (unless you select to show it).

Root password:

••••••••

☐ Show Password in Clear

Please enter the same root password again to verify that you have typed it correctly.

Re-enter password to verify:

••••••••

☐ Show Password in Clear



Set up users and passwords

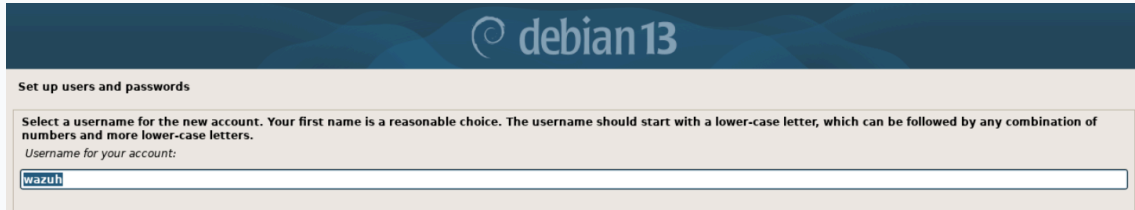
A user account will be created for you to use instead of the root account for non-administrative activities.

Please enter the real name of this user. This information will be used for instance as default origin for emails sent by this user as well as any program which displays or uses the user's real name. Your full name is a reasonable choice.

Full name for the new user:

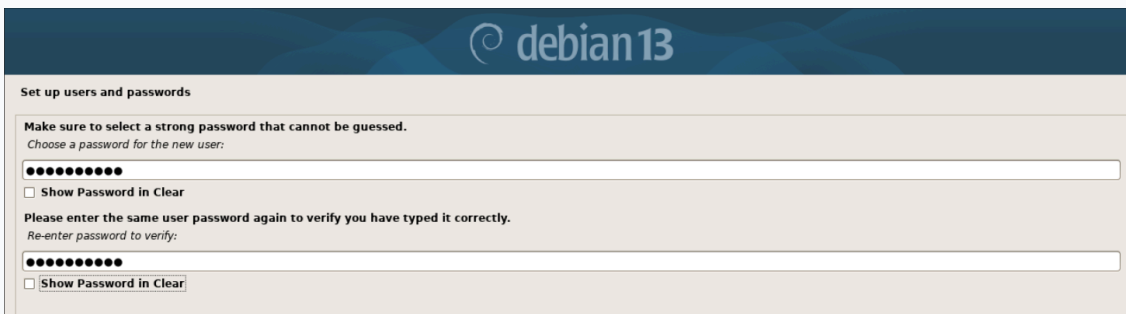
Wazuh ITELECT3

For security and access, create a user with the username wazuh. For both the administrative



The screenshot shows the 'Set up users and passwords' screen in the Debian 13 installer. The header features the Debian 13 logo. Below the header, the title 'Set up users and passwords' is displayed. The main text instructs the user to 'Select a username for the new account. Your first name is a reasonable choice. The username should start with a lower-case letter, which can be followed by any combination of numbers and more lower-case letters.' Below this, a label 'Username for your account:' is followed by a text input field containing the username 'wazuh'.

root and this user account, apply the password [REDACTED FOR PUBLIC REPOSITORY].



The screenshot shows the 'Set up users and passwords' screen in the Debian 13 installer, specifically the password setup section. The header features the Debian 13 logo. Below the header, the title 'Set up users and passwords' is displayed. The main text instructs the user to 'Make sure to select a strong password that cannot be guessed.' and 'Choose a password for the new user:'. Below this, there is a password input field with a 'Show Password in Clear' checkbox. The next section instructs the user to 'Please enter the same user password again to verify you have typed it correctly.' and 'Re-enter password to verify:'. Below this, there is a second password input field with a 'Show Password in Clear' checkbox.

When prompted for the disk configuration, select Use entire disk to allow the system to manage the available storage. Follow the SCSI1 (0,0,0) (sda) - 53.7 GB QEMU QEMU HARDDISK path;



The screenshot shows the 'Partition disks' screen in the Debian 13 installer. The header features the Debian 13 logo. Below the header, the title 'Partition disks' is displayed. The main text instructs the user to 'The installer can guide you through partitioning a disk (using different standard schemes) or, if you prefer, you can do it manually. With guided partitioning you will still have a chance later to review and customise the results.' Below this, a label 'Partitioning method:' is followed by a list of options: 'Guided - use entire disk', 'Guided - use entire disk and set up LVM', 'Guided - use entire disk and set up encrypted LVM', and 'Manual'. The 'Guided - use entire disk' option is selected. Below this, the text 'Note that all data on the disk you select will be erased, but not before you have confirmed that you really want to make the changes.' is displayed. Below this, a label 'Select disk to partition:' is followed by a list of disks: 'SCSI1 (0,0,0) (sda) - 53.7 GB QEMU QEMU HARDDISK'. This disk is selected.

And choose the option to place All files in one partition—this is ideal for new users and standard lab setups. Finally, select Finish partitioning



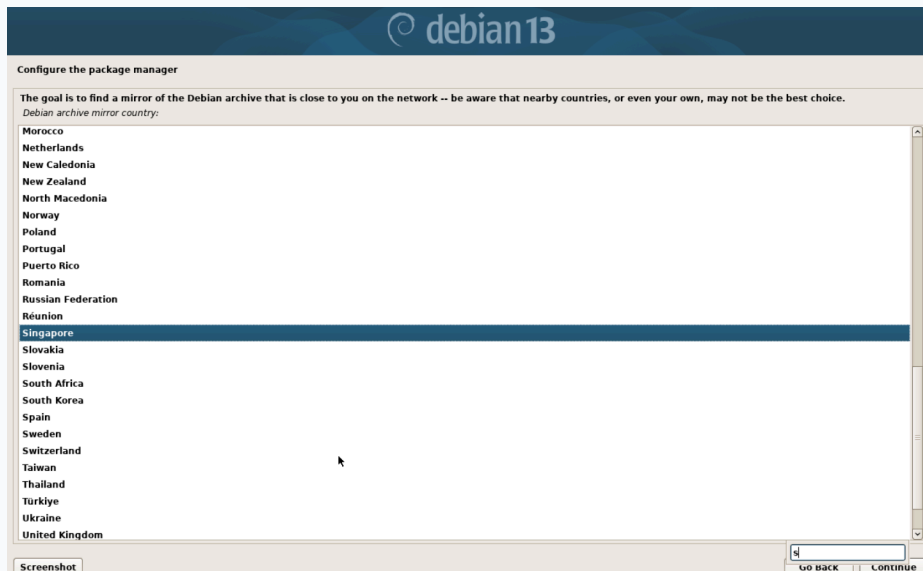
After that we will choose YES to write changes to the disk to begin the installation.



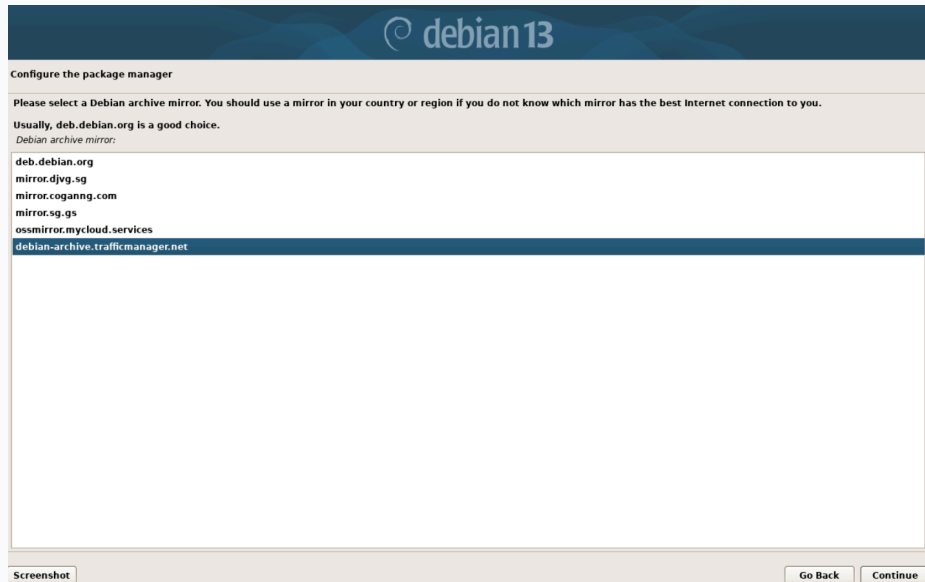
After that, it will ask you if you want to scan extra installation media. For this part, we will choose the NO



When configuring the package manager, select Singapore as the Debian archive mirror country. For the specific mirror address, choose `debian-archive.traffikmanager.net`.







For the remaining package manager settings, such as HTTP proxy configurations, leave all fields at their default values without making any changes. Simply click Continue

For the remaining package manager settings, such as HTTP proxy configurations, leave all fields at their default values without making any changes. Simply click Continue

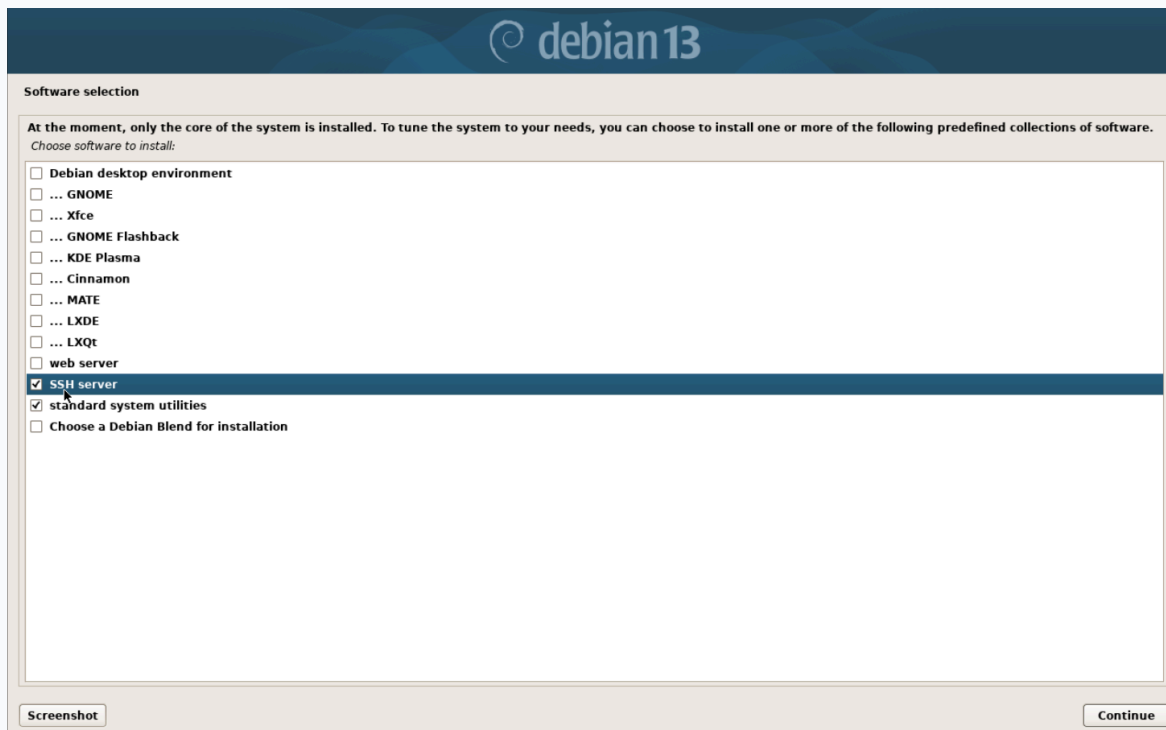




When the installer prompts you for the package usage survey (popularity-contest), select No.



On the software selection screen, ensure that only SSH Server and Standard System Utilities are checked.



The image shows the 'Software selection' window from the Debian 13 installer. The window has a dark blue header with the 'debian 13' logo. Below the header, the title 'Software selection' is displayed. The main content area contains a list of software collections to install, each with a checkbox. The following table represents the visible options:

| Software Collection                    | Selected                            |
|----------------------------------------|-------------------------------------|
| Debian desktop environment             | <input type="checkbox"/>            |
| ... GNOME                              | <input type="checkbox"/>            |
| ... Xfce                               | <input type="checkbox"/>            |
| ... GNOME Flashback                    | <input type="checkbox"/>            |
| ... KDE Plasma                         | <input type="checkbox"/>            |
| ... Cinnamon                           | <input type="checkbox"/>            |
| ... MATE                               | <input type="checkbox"/>            |
| ... LXDE                               | <input type="checkbox"/>            |
| ... LXQt                               | <input type="checkbox"/>            |
| web server                             | <input type="checkbox"/>            |
| SSH server                             | <input checked="" type="checkbox"/> |
| standard system utilities              | <input checked="" type="checkbox"/> |
| Choose a Debian Blend for installation | <input type="checkbox"/>            |

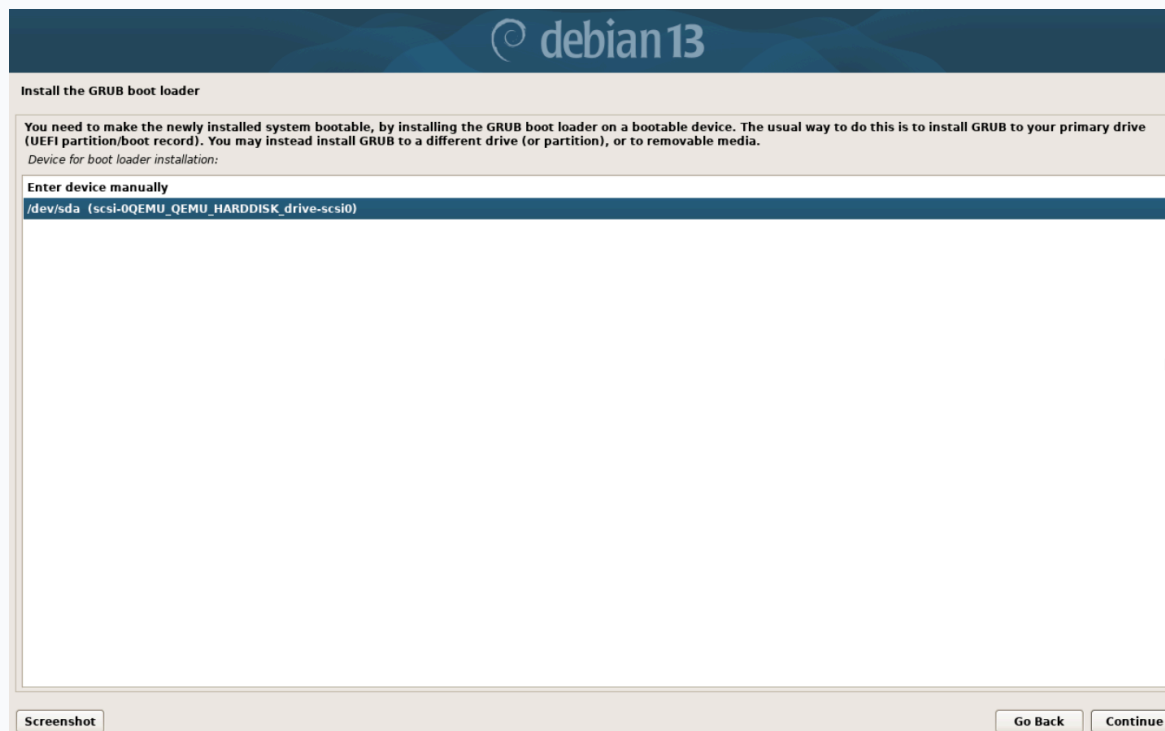
At the bottom of the window, there are two buttons: 'Screenshot' on the left and 'Continue' on the right.



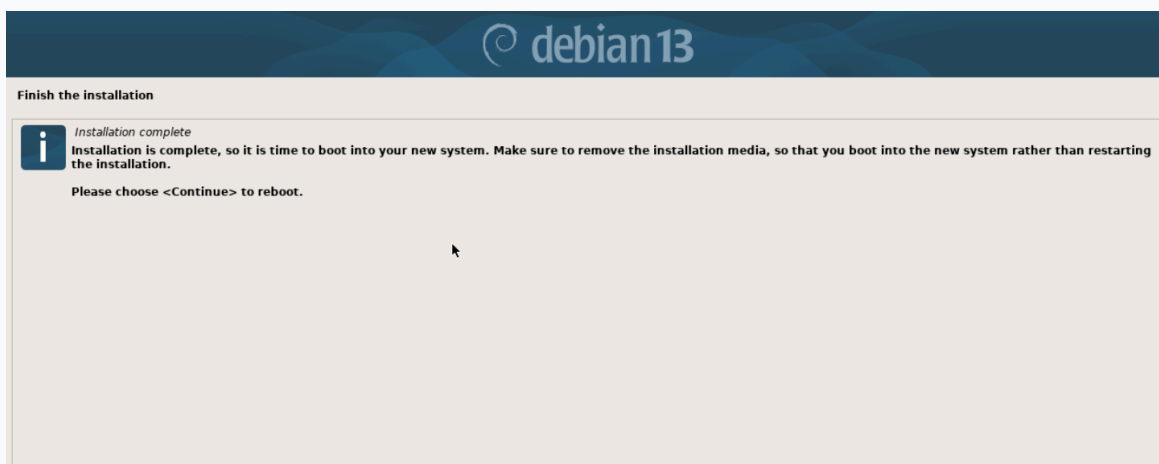
When prompted to install the GRUB boot loader to your primary drive, select Yes.



When asked where to install the boot loader, select the local disk path—typically `/dev/sda` (`scsi-0QEMU_QEMU_HARDDISK_drive-scsi0`)



Once you see the "Installation Complete" message, click Continue. The VM will reboot, and you will be greeted by the login prompt where you can enter the credentials we set up earlier.



## INSTALLATION OF TAILSCALE INSIDE WAZUH

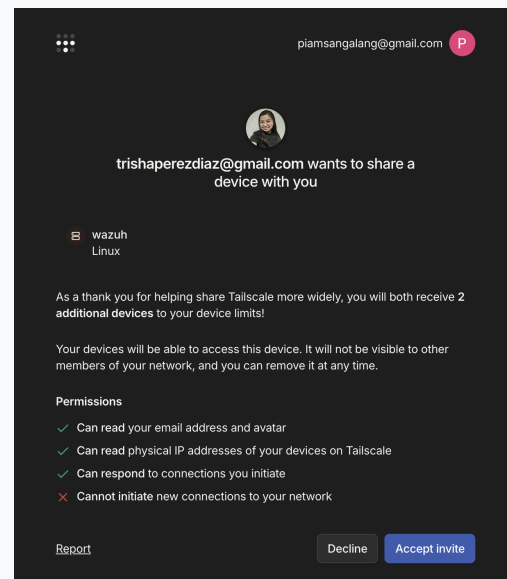
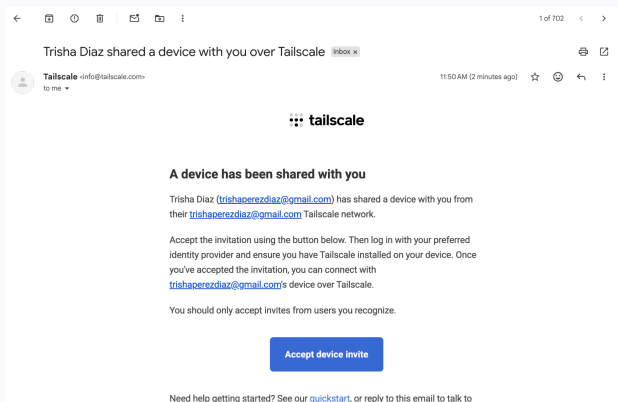
wazuh login : wazuh

Password : [REDACTED FOR PUBLIC REPOSITORY]

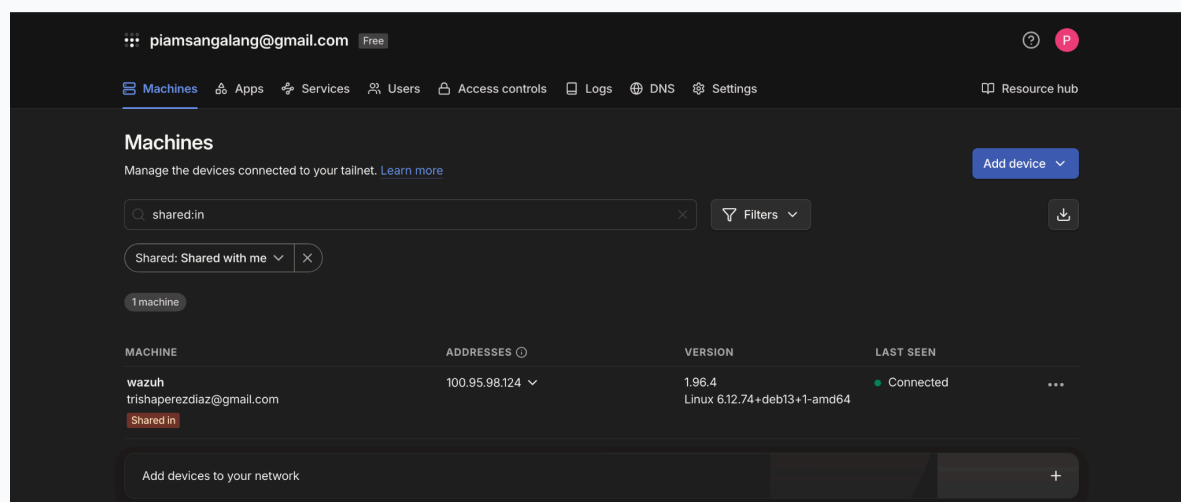
wazuh@wazuh:~\$ su -

## CONNECTING TO OTHER USERS

Share the Device on Tailscale Website - <https://login.tailscale.com/admin/machines>



Once the recipient opens their email, the invitation will be visible in the main inbox. If it isn't immediately apparent, we recommend checking the 'Spam' folders. Just click *Accept Invite*



Once you click Accept Invite, you will be redirected to your Tailscale admin dashboard. There, you'll see the shared machine appearing instantly in your tailnet list, marked with a 'Shared In' tag to distinguish it from your own devices.

## Agent Deployment

Wazuh Agents collect security data from monitored endpoints and forward it to the Manager. Agents can be deployed from within the Dashboard using a simple guided wizard:

1. In the Wazuh Dashboard, navigate to the Agents section and click Deploy new agent.
2. Select the operating system of the target endpoint (Linux, Windows, macOS, etc.).
3. Configure the Manager IP address (the cloud VM IP).
4. Copy the generated installation command and run it on the target machine via SSH or terminal.
5. The agent will auto-register with the Manager and appear under Agents Summary as Active.

# DEPLOYING AGENTS (KALI)

W. Endpoints Deploy new agent

Deploy new agent

LINUX

☐ RPM amd64

☒ DEB amd64

☐ RPM aarch64

☐ DEB aarch64

WINDOWS

☐ MSI 32/64 bits

macOS

☐ Intel

☐ Apple silicon

For additional systems and architectures, please check our documentation.

Server address

This is the address the agent uses to communicate with the server. Enter an IP address or a fully qualified domain name (FQDN).

Assign a server address

192.168.1.101

☒ Remember server address

Optional settings

By default, the deployment uses the hostname as the agent name. Optionally, you can use a different agent name in the field below.

Assign an agent name

kali\_trisha

W. Endpoints Deploy new agent

Assign an agent name

kali\_trisha

The agent name must be unique. It can't be changed once the agent has been enrolled.

Select one or more existing groups

Default

Run the following commands to download and install the agent:

wget https://packages.wazuh.com/4.x/spt/pool/main/w/wazuh-agent/wazuh-agent\_4.10.1-1\_amd64.deb && sudo WAZUH\_MANAGER='192.168.1.101' WAZUH\_AGENT\_NAME='kali\_trisha' dpkg -i ./wazuh-agent\_4.10.1-1\_amd64.deb

Requirements

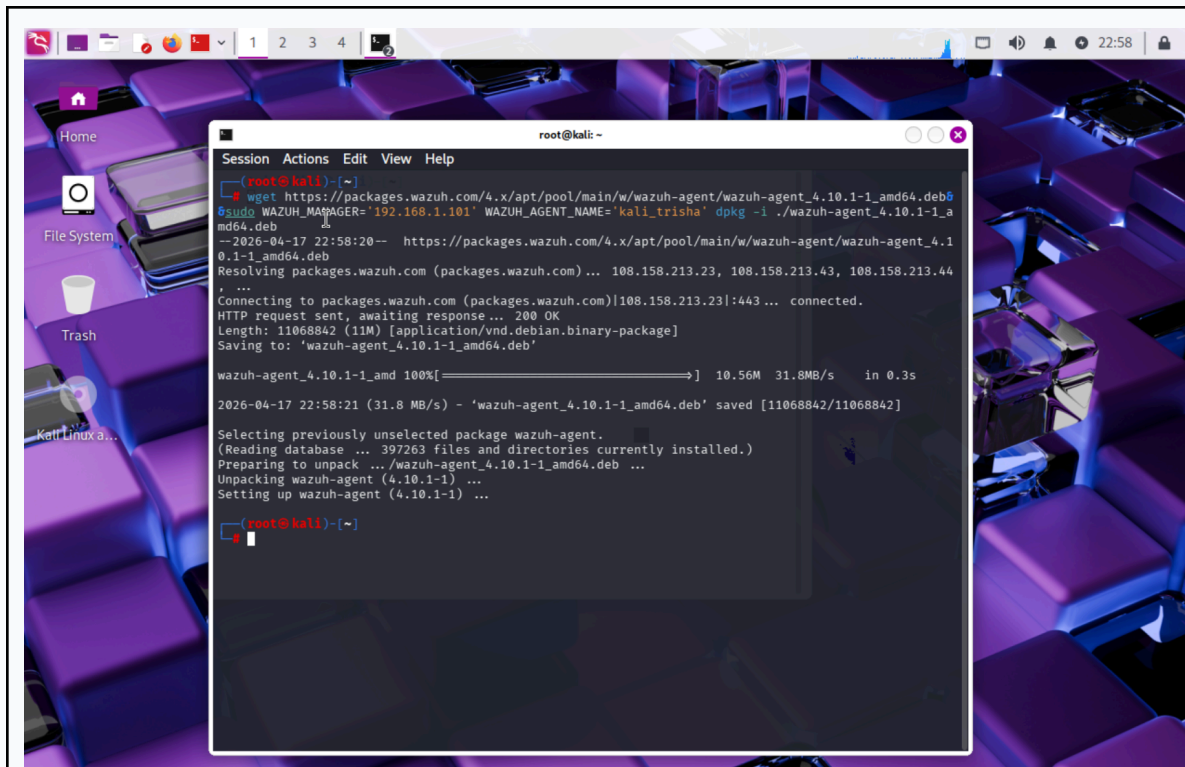
- You will need administrator privileges to perform this installation.
- Shell Bash is required.

Keep in mind you need to run this command in a Shell Bash terminal.

Start the agent:

sudo systemctl daemon-reload  
sudo systemctl enable wazuh-agent  
sudo systemctl start wazuh-agent





```
(root@kali)-[~]
# sudo systemctl daemon-reload

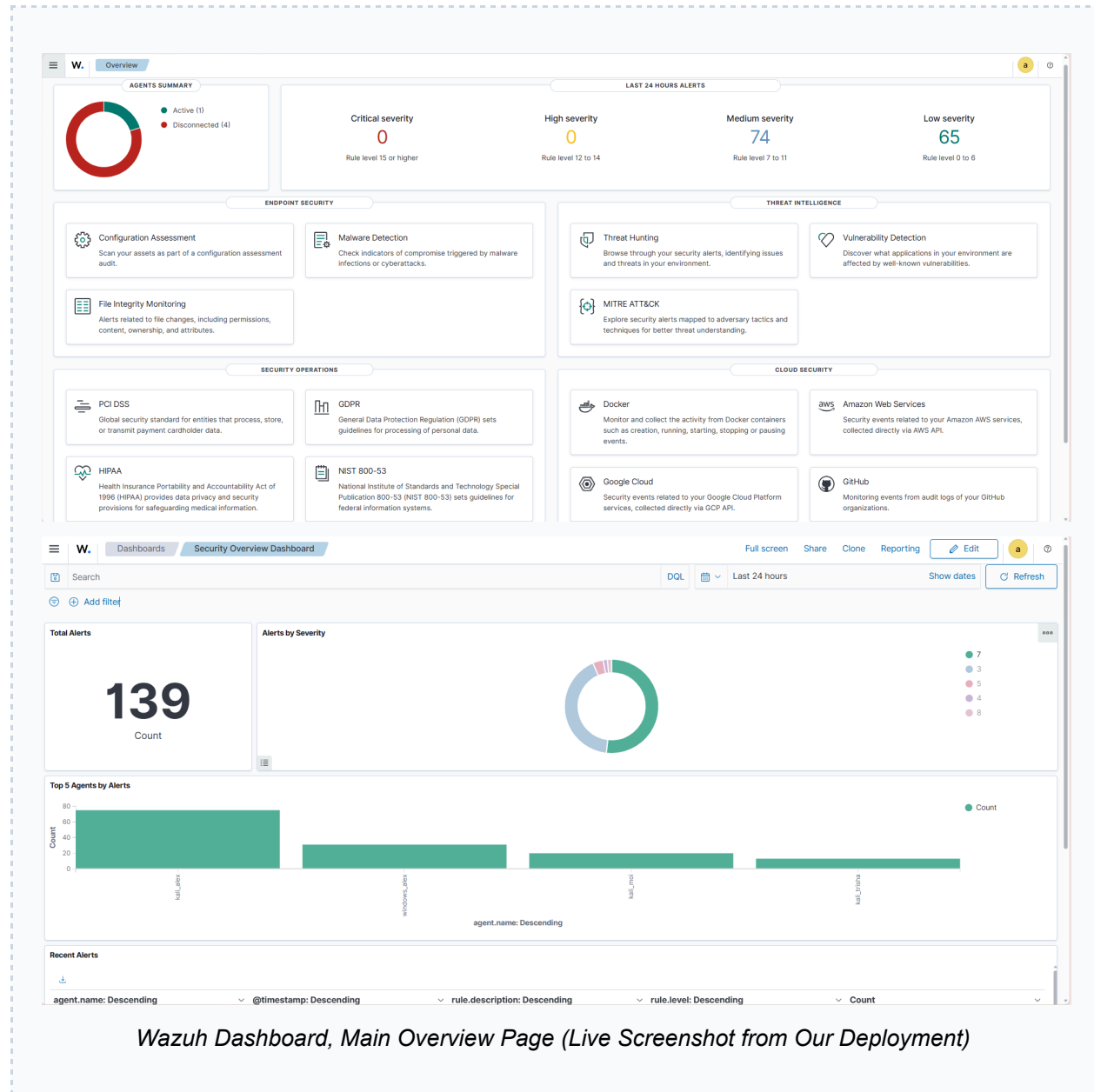
(root@kali)-[~]
# sudo systemctl enable wazuh-agent
Created symlink '/etc/systemd/system/multi-user.target.wants/wazuh-agent.service' → '/usr/lib/systemd/system/wazuh-agent.service'.

(root@kali)-[~]
# sudo systemctl start wazuh-agent

(root@kali)-[~]
#
```

# Dashboard Overview

The Wazuh Dashboard serves as the central command interface for the entire security platform. It delivers real-time visibility into security events, agent health, threat intelligence, and regulatory compliance, all from a single, unified web interface.



Wazuh Dashboard, Main Overview Page (Live Screenshot from Our Deployment)

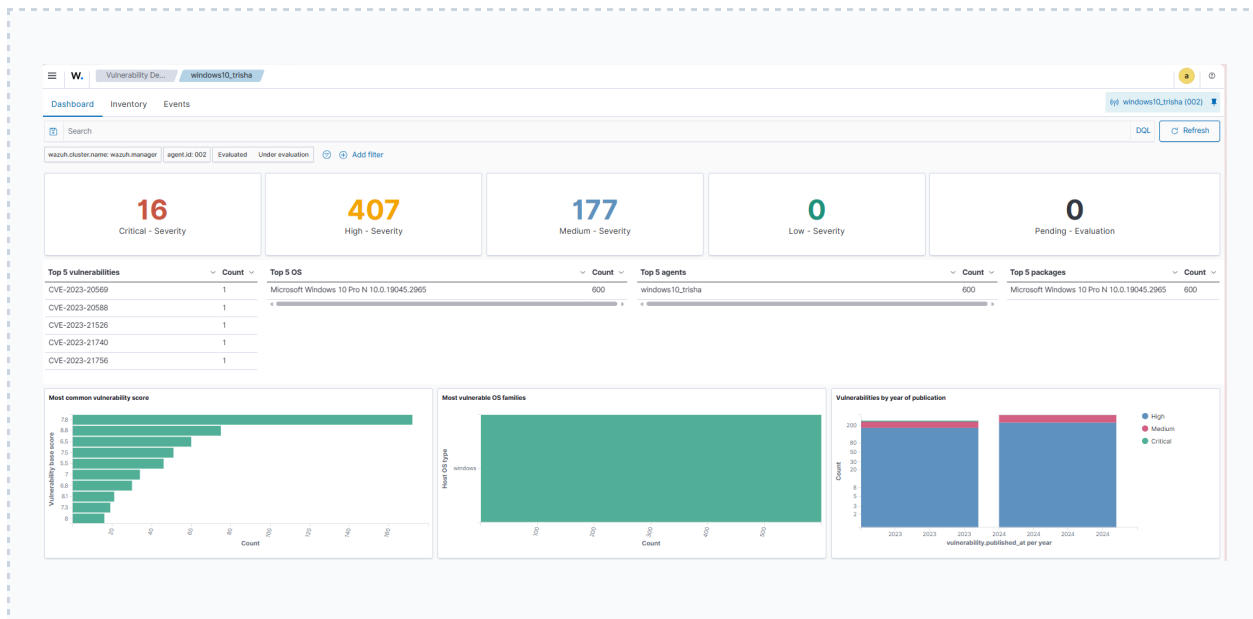
# Features & Functionality

Wazuh organizes its capabilities into four major security domains: Endpoint Security, Threat Intelligence, Security Operations, and Cloud Security. This section details each feature, its purpose, how it works, and its relevance to ethical hacking and security assessment.

## Vulnerability Detection

Vulnerability Detection is Wazuh's automated process for identifying known Common Vulnerabilities and Exposures (CVEs) across all monitored endpoints. Wazuh maintains an up-to-date vulnerability database and continuously correlates installed software against known vulnerability feeds.

- The Wazuh Agent collects a complete software inventory from the endpoint, including installed packages, versions, and paths.
- The Wazuh Manager correlates the inventory against the National Vulnerability Database (NVD), the Red Hat Security Database, Debian/Ubuntu CVE feeds, and vendor-specific advisories.
- Matched vulnerabilities are classified by severity (Critical, High, Medium, Low) using their CVSS score and displayed on the Dashboard.
- Each finding includes: CVE ID, CVSS score, affected package, version, and recommended remediation steps.



W. Vulnerability De... windows10\_trisha

DashboardInventoryEvents

Search

wazuh.cluster.name: wazuh.manageragent.id: 002EvaluatedUnder evaluationAdd filter

600 hits

Export Formatted46 available fieldsColumnsDensitySort fieldsFull screen

| agent.name       | package.name                               | package.ver...  | vulnerability.description                                                          | vulnerability.severity | vulnerability.id |
|------------------|--------------------------------------------|-----------------|------------------------------------------------------------------------------------|------------------------|------------------|
| windows10_trisha | Microsoft Windows 10 Pro N 10.0.19045.2965 | 10.0.19045.2965 | Windows USB Video Class System Driver Elevation of Privilege Vulnerability         | Medium                 | CVE-2024-43643   |
| windows10_trisha | Microsoft Windows 10 Pro N 10.0.19045.2965 | 10.0.19045.2965 | Windows USB Video Class System Driver Elevation of Privilege Vulnerability         | Medium                 | CVE-2024-43634   |
| windows10_trisha | Microsoft Windows 10 Pro N 10.0.19045.2965 | 10.0.19045.2965 | Windows Kernel Elevation of Privilege Vulnerability                                | High                   | CVE-2024-43630   |
| windows10_trisha | Microsoft Windows 10 Pro N 10.0.19045.2965 | 10.0.19045.2965 | Windows DWM Core Library Elevation of Privilege Vulnerability                      | High                   | CVE-2024-43629   |
| windows10_trisha | Microsoft Windows 10 Pro N 10.0.19045.2965 | 10.0.19045.2965 | Windows Telephony Service Remote Code Execution Vulnerability                      | High                   | CVE-2024-43628   |
| windows10_trisha | Microsoft Windows 10 Pro N 10.0.19045.2965 | 10.0.19045.2965 | Windows Telephony Service Elevation of Privilege Vulnerability                     | High                   | CVE-2024-43626   |
| windows10_trisha | Microsoft Windows 10 Pro N 10.0.19045.2965 | 10.0.19045.2965 | Remote Desktop Client Remote Code Execution Vulnerability                          | High                   | CVE-2024-43599   |
| windows10_trisha | Microsoft Windows 10 Pro N 10.0.19045.2965 | 10.0.19045.2965 | Code Integrity Guard Security Feature Bypass Vulnerability                         | Medium                 | CVE-2024-43585   |
| windows10_trisha | Microsoft Windows 10 Pro N 10.0.19045.2965 | 10.0.19045.2965 | Remote Desktop Protocol Server Remote Code Execution Vulnerability                 | High                   | CVE-2024-43582   |
| windows10_trisha | Microsoft Windows 10 Pro N 10.0.19045.2965 | 10.0.19045.2965 | Microsoft Speech Application Programming Interface (SAPI) Remote Code Executi...   | High                   | CVE-2024-43574   |
| windows10_trisha | Microsoft Windows 10 Pro N 10.0.19045.2965 | 10.0.19045.2965 | Windows MSHTML Platform Spoofing Vulnerability                                     | Medium                 | CVE-2024-43573   |
| windows10_trisha | Microsoft Windows 10 Pro N 10.0.19045.2965 | 10.0.19045.2965 | Windows Network Address Translation (NAT) Denial of Service Vulnerability          | High                   | CVE-2024-43565   |
| windows10_trisha | Microsoft Windows 10 Pro N 10.0.19045.2965 | 10.0.19045.2965 | Microsoft Management Console Remote Code Execution Vulnerability                   | High                   | CVE-2024-43572   |
| windows10_trisha | Microsoft Windows 10 Pro N 10.0.19045.2965 | 10.0.19045.2965 | Windows Ancillary Function Driver for WinSock Elevation of Privilege Vulnerability | High                   | CVE-2024-43563   |
| windows10_trisha | Microsoft Windows 10 Pro N 10.0.19045.2965 | 10.0.19045.2965 | Windows Mobile Broadband Driver Denial of Service Vulnerability                    | Medium                 | CVE-2024-43558   |

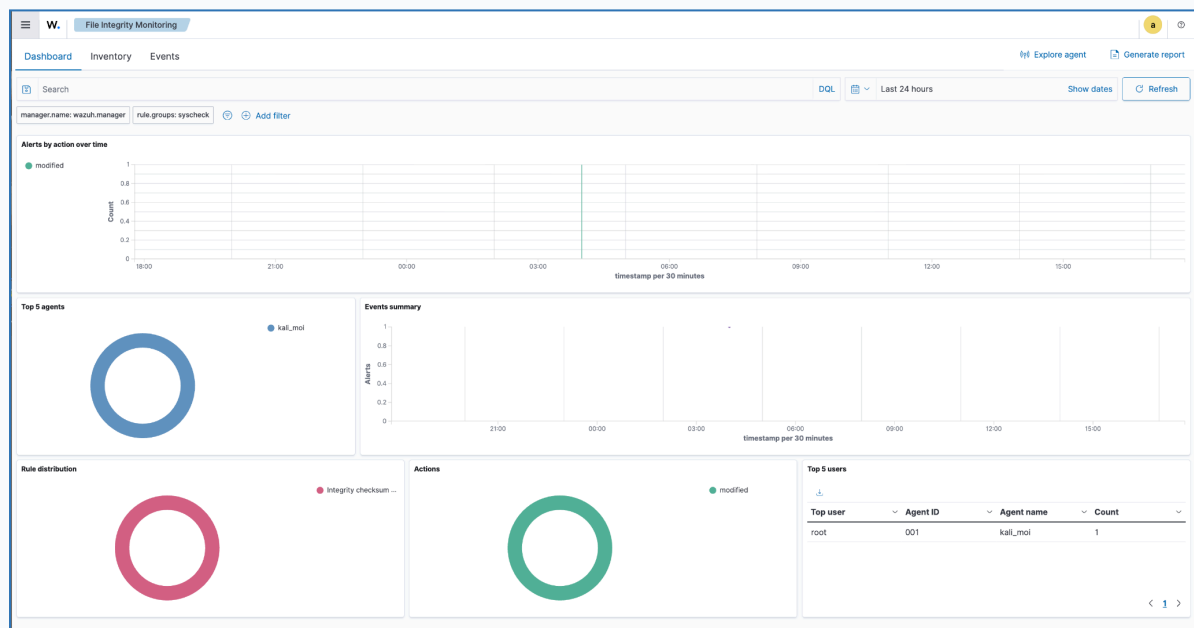
Rows per page: 15

Vulnerability Detection, CVE Findings List by Severity

## File Integrity Monitoring (FIM)

File Integrity Monitoring (FIM) is one of Wazuh's most powerful real-time capabilities. It continuously watches specified files and directories for unauthorized changes, generating cryptographic checksums (MD5, SHA1, SHA256) and alerting on any modification to content, permissions, ownership, or attributes.

- Detects file creation, modification, and deletion in real time.
- Monitors file permissions, ownership, and attribute changes.
- Records before/after checksums for forensic comparison.
- Configurable scope: system files, config directories, web roots, custom paths.
- Works on Windows (registry monitoring) and Linux/macOS.



*File Integrity Monitoring, Alert with Before/After Hash Comparison*

## Log Data Analysis

The Wazuh Log Analysis engine is a high-performance, rule-based system that continuously collects, normalizes, parses, and analyzes logs from hundreds of sources in real time. With over 3,000 built-in rules, it detects known attack signatures, anomalous patterns, and policy violations.

W.

Logs

a

Logs

[Export formatted](#)

List and filter logs.

All daemons

All log levels

☐ Descending sort

☐ Realtime

Filter logs

May 1, 2026 @ 08:00:10.000 wazuh-monitord INFO Starting new log after rotation.

May 1, 2026 @ 08:34:01.000 wazuh-modulesd:content-updater WARNING Offset processing failed. Triggered a snapshot download.

May 1, 2026 @ 08:34:01.000 wazuh-modulesd:content-updater WARNING Couldn't run full content download: Error -1 from server: Could not resolve hostname.

May 1, 2026 @ 08:34:01.000 wazuh-modulesd:content-updater ERROR Action for 'vulnerability\_feed\_manager' failed: Error -1 from server: Could not resolve hostname.

May 1, 2026 @ 08:34:49.000 wazuh-modulesd:syscollector INFO Starting evaluation.

May 1, 2026 @ 08:34:51.000 wazuh-modulesd:syscollector INFO Evaluation finished.

May 1, 2026 @ 09:34:01.000 wazuh-modulesd:content-updater WARNING Offset processing failed. Triggered a snapshot download.

May 1, 2026 @ 09:34:01.000 wazuh-modulesd:content-updater WARNING Couldn't run full content download: Error -1 from server: Could not resolve hostname.

May 1, 2026 @ 09:34:01.000 wazuh-modulesd:content-updater ERROR Action for 'vulnerability\_feed\_manager' failed: Error -1 from server: Could not resolve hostname.

May 1, 2026 @ 09:34:52.000 wazuh-modulesd:syscollector INFO Starting evaluation.

May 1, 2026 @ 09:34:54.000 wazuh-modulesd:syscollector INFO Evaluation finished.

May 1, 2026 @ 10:34:01.000 wazuh-modulesd:content-updater WARNING Offset processing failed. Triggered a snapshot download.

May 1, 2026 @ 10:34:01.000 wazuh-modulesd:content-updater WARNING Couldn't run full content download: Error -1 from server: Could not resolve hostname.

May 1, 2026 @ 10:34:01.000 wazuh-modulesd:content-updater ERROR Action for 'vulnerability\_feed\_manager' failed: Error -1 from server: Could not resolve hostname.

May 1, 2026 @ 10:34:55.000 wazuh-modulesd:syscollector INFO Starting evaluation.

May 1, 2026 @ 10:34:56.000 wazuh-modulesd:syscollector INFO Evaluation finished.

May 1, 2026 @ 11:34:01.000 wazuh-modulesd:content-updater WARNING Offset processing failed. Triggered a snapshot download.

May 1, 2026 @ 11:34:01.000 wazuh-modulesd:content-updater WARNING Couldn't run full content download: Error -1 from server: Could not resolve hostname.

May 1, 2026 @ 11:34:01.000 wazuh-modulesd:content-updater ERROR Action for 'vulnerability\_feed\_manager' failed: Error -1 from server: Could not resolve hostname.

May 1, 2026 @ 11:34:57.000 wazuh-modulesd:syscollector INFO Starting evaluation.

May 1, 2026 @ 11:34:59.000 wazuh-modulesd:syscollector INFO Evaluation finished.

May 1, 2026 @ 12:30:24.000 wazuh-db INFO Created Global database backup: "backup/db/global.db-backup-2026-05-01-04:30:24.gz"

May 1, 2026 @ 12:30:24.000 wazuh-db INFO Deleted Global database backup: "backup/db/global.db-backup-2026-04-19-05:54:23.gz"

May 1, 2026 @ 12:34:01.000 wazuh-modulesd:content-updater WARNING Offset processing failed. Triggered a snapshot download.

May 1, 2026 @ 12:34:01.000 wazuh-modulesd:content-updater WARNING Couldn't run full content download: Error -1 from server: Could not resolve hostname.

May 1, 2026 @ 12:34:01.000 wazuh-modulesd:content-updater ERROR Action for 'vulnerability\_feed\_manager' failed: Error -1 from server: Could not resolve hostname.

May 1, 2026 @ 12:35:00.000 wazuh-modulesd:syscollector INFO Starting evaluation.

May 1, 2026 @ 12:35:02.000 wazuh-modulesd:syscollector INFO Evaluation finished.

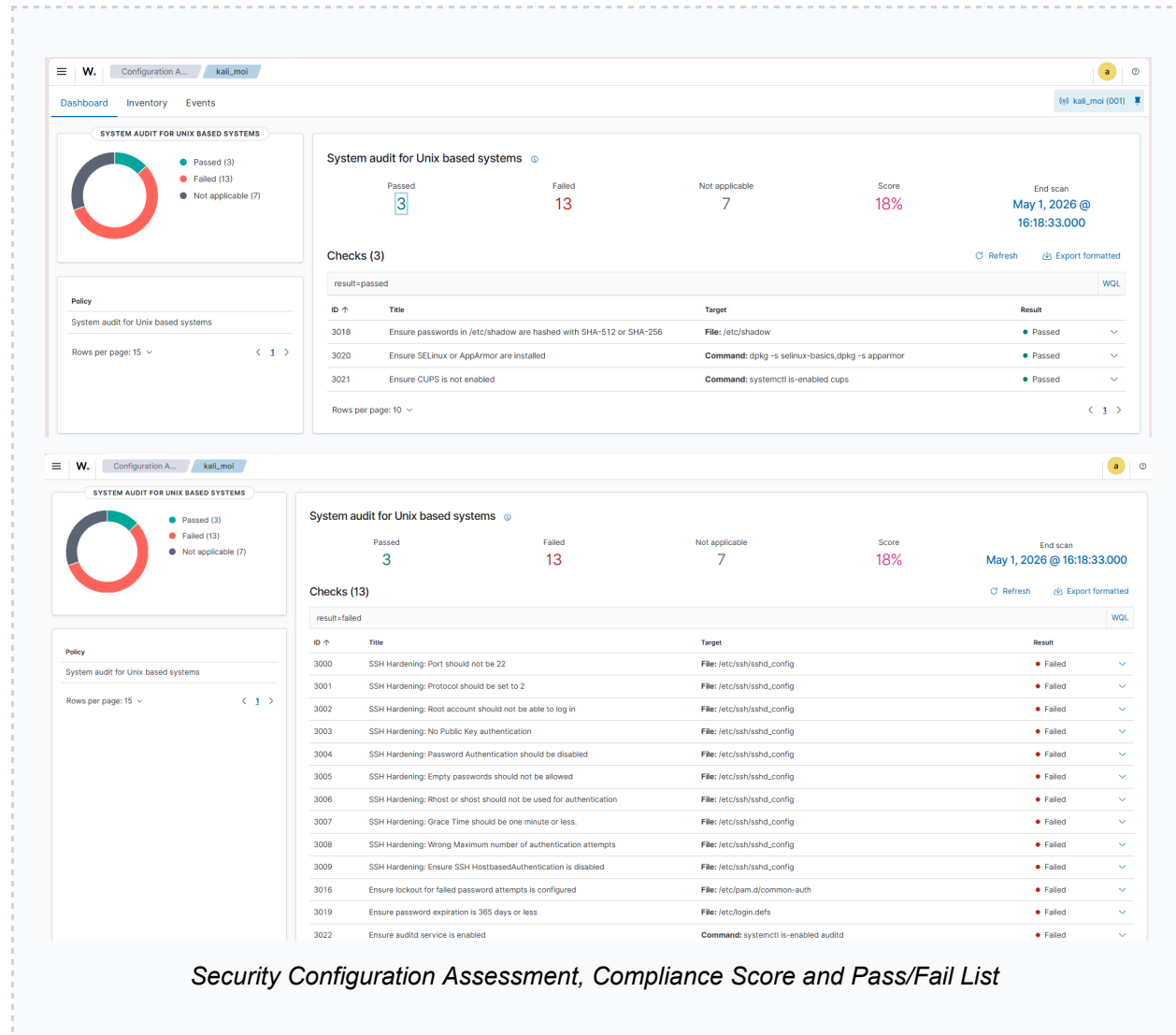
May 1, 2026 @ 13:34:01.000 wazuh-modulesd:content-updater WARNING Offset processing failed. Triggered a snapshot download.

May 1, 2026 @ 13:34:01.000 wazuh-modulesd:content-updater WARNING Couldn't run full content download: Error -1 from server: Could not resolve

Log Analysis, Security Events Timeline View

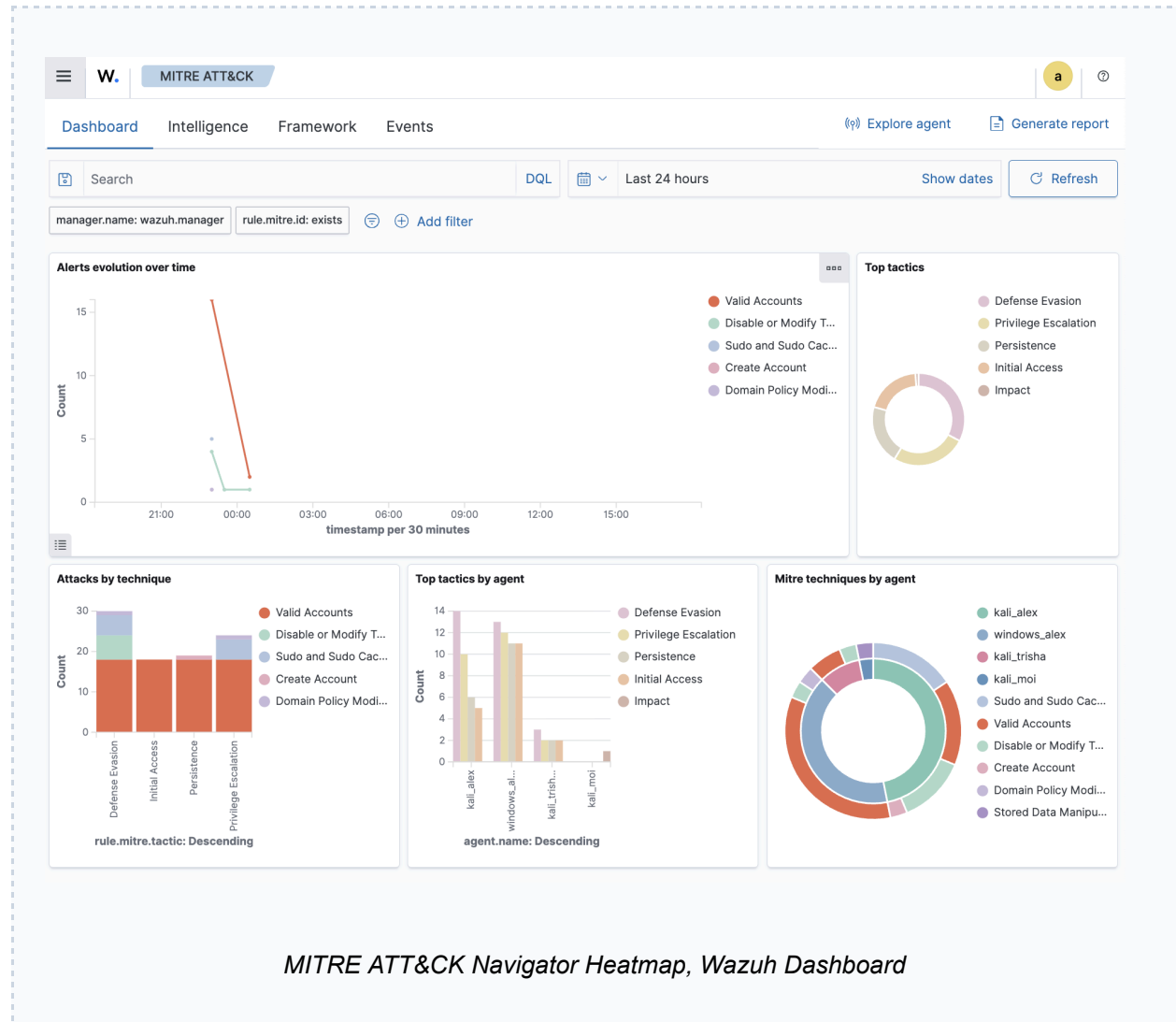
## Security Configuration Assessment (SCA)

SCA automatically audits the configuration of monitored endpoints against industry-accepted security benchmarks. It identifies misconfigurations that could be exploited, such as weak SSH settings, open services, disabled firewalls, and insecure file permissions, then produces a compliance score with actionable remediation guidance.



## MITRE ATT&CK Integration

Wazuh maps all detected security alerts to the MITRE ATT&CK framework, the globally recognized knowledge base of adversary tactics, techniques, and procedures (TTPs). This enriches every alert with strategic context, helping analysts understand not just what happened, but what an attacker was trying to achieve.





# Demonstration Scenarios

All demonstrations were conducted in our live Wazuh v4.10.1 environment. No external systems were harmed. All simulated attacks were performed on our own controlled endpoints for educational purposes only.

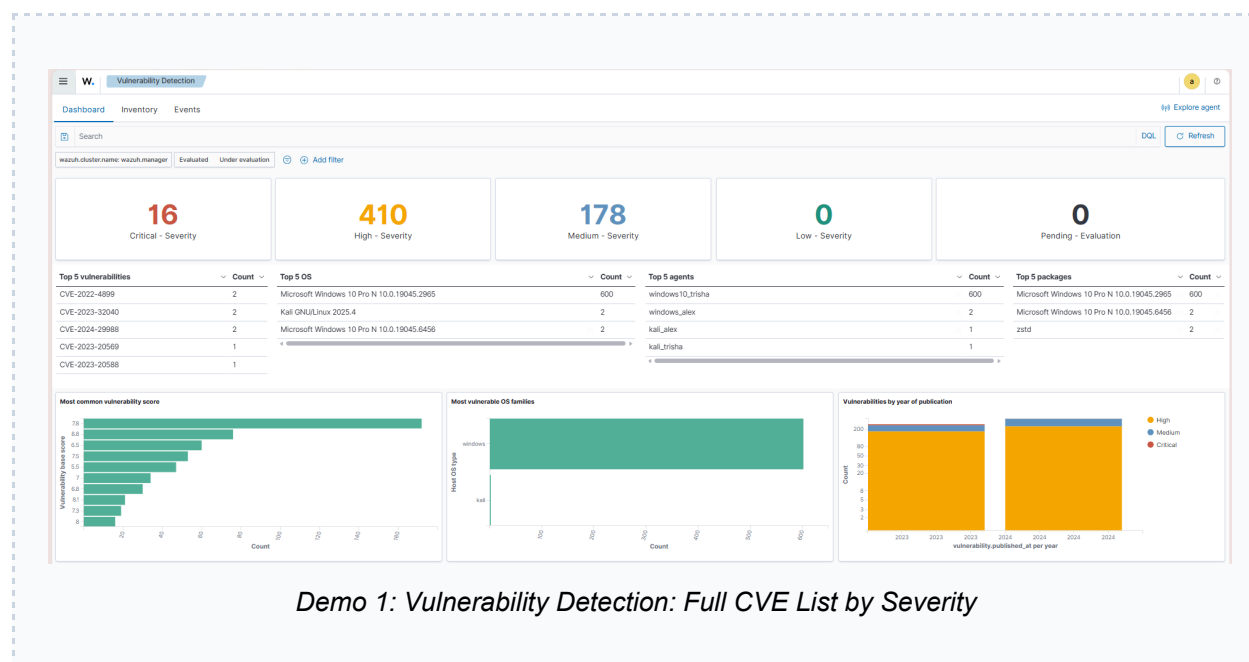
## Demo 1: Vulnerability Detection Scan

### Objective

Show how Wazuh automatically discovers and classifies known CVEs present in the software installed on monitored endpoints , without any manual scanning commands.

### Demonstration Steps

1. Navigate to Threat Intelligence > Vulnerability Detection on the Dashboard.
2. Select a connected agent from the agent list to view its specific vulnerability findings.
3. Review the CVE list filtered by severity (Critical, High, Medium, Low).
4. Click a specific CVE to view: CVE ID, CVSS score, affected package, version, and remediation.



### Results Discussion

The vulnerability scan across monitored endpoints revealed a substantial number of unpatched CVEs, with agent windows10\_trisha accounting for the majority of findings, 16 Critical and 410 High severity vulnerabilities across 600 scanned packages. This volume indicates a significantly outdated Windows environment with a wide exploitable attack surface.

From an ethical hacking perspective, Critical and High severity CVEs with publicly available exploits would be the first targets during an exploitation phase, as they offer the highest probability of successful compromise with minimal effort. The fact that Wazuh detected and classified all of these automatically, without a single manual scanning command, demonstrates its value as a continuous vulnerability management tool, providing defenders with an always-current picture of endpoint exposure rather than a snapshot from periodic scans.

## Demo 2: Malware Detection via Rootcheck

### Objective

Demonstrate how Wazuh's rootcheck module automatically detects trojaned system binaries on monitored endpoints, identifying signs of active malware or post-exploitation activity without any manual scanning commands.

### Demonstration Steps

1. Navigate to Explore → Discover on the Dashboard.
2. In the search bar, enter rule.groups: rootcheck and set the time range to Last 7 days.
3. Review the results — 60 hits appear, all originating from agent kali\_moi (192.168.1.97).
4. Observe the recurring data.title: Trojaned version of file detected across multiple entries.
5. Note the unique files flagged across all alerts: /usr/bin/chsh, /bin/chsh, /usr/bin/chfn, /bin/chfn, /usr/bin/passwd, and /bin/passwd.
6. Click to expand one alert and review the full forensic detail in the Table view.



Demo 2: 60 Rootcheck Hits Showing Trojaned File Detections on kali\_moi

May 1, 2026 16:28:53.245

input.type: log agent.ip: 192.168.1.97 agent.name: kali\_moi agent.id: 001 manager.name: wazuh.manager data.file: /usr/bin/chsh data.title: Trojaned version of file detected.  
 rule.firedtimes: 6 rule.mitre: false rule.level: 7 rule.pci.dss: 16.6.1 rule.description: Host-based anomaly detection event (rootcheck). rule.groups: ossec, rootcheck rule.id: 518  
 rule.gdpr: IV.35.7.d location: rootcheck decoder.name: rootcheck id: 1777623653.5797 full\_log: Trojaned version of file '/usr/bin/chsh' detected. Signature used: 'bash(file.hiproc.h)/dev/ttyo/dev/[A-Z]/dev/[a-s,uxxz]' (Generic). timestamp: May 1, 2026 @ 16:28:53.245 \_index: wazuh-alerts-4.x-2026.05.01

Expanded document

Table JSON

|                  |                                                                                                                                       |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| _index           | wazuh-alerts-4.x-2026.05.01                                                                                                           |
| agent.id         | 001                                                                                                                                   |
| agent.ip         | 192.168.1.97                                                                                                                          |
| agent.name       | kali_moi                                                                                                                              |
| data.file        | /usr/bin/chsh                                                                                                                         |
| data.title       | Trojaned version of file detected.                                                                                                    |
| decoder.name     | rootcheck                                                                                                                             |
| full_log         | Trojaned version of file '/usr/bin/chsh' detected. Signature used: 'bash(file.hiproc.h)/dev/ttyo/dev/[A-Z]/dev/[a-s,uxxz]' (Generic). |
| id               | 1777623653.5797                                                                                                                       |
| input.type       | log                                                                                                                                   |
| location         | rootcheck                                                                                                                             |
| manager.name     | wazuh.manager                                                                                                                         |
| rule.description | Host-based anomaly detection event (rootcheck).                                                                                       |
| rule.firedtimes  | 6                                                                                                                                     |

*Expanded Alert Detail: Trojaned Binary Fields Including File Path, Rule, and Compliance Tags*

## Results Discussion

The rootcheck module flagged six trojaned system binaries on kali\_moi across 60 alerts over 7 days: /usr/bin/chsh, /bin/chsh, /usr/bin/chfn, /bin/chfn, /usr/bin/passwd, and /bin/passwd. These are core Linux utilities responsible for managing user credentials and shell access, precisely the binaries an attacker would replace after gaining root access to establish persistence and harvest credentials. The presence of trojaned versions in both /usr/bin and /bin paths, combined with the rule firing repeatedly across multiple scan cycles, is consistent with a rootkit installation rather than an isolated incident. From an ethical hacking perspective, this is the most critical finding in the entire deployment, indicating the endpoint was not merely vulnerable but actively compromised at the system level.

## Demo 3: File Integrity Monitoring (FIM)

### Objective

Demonstrate how Wazuh instantly detects unauthorized modifications to critical system files , simulating what a real attacker might do after gaining initial access to a compromised system.

### Demonstration Steps

1. Navigate to the File Integrity Monitoring module.
2. Review the Dashboard tab → observer active users and actions
3. Check the Events chart → spot the modification spike
4. Note files modified vs. added vs. deleted

5. Switch to Inventory Tab → browse monitored files
6. Switch to Events Tab → examine alert spikes
7. Examine Events Field → confirm the triggered rule

W. File Integrity M. kal\_mol

Dashboard Inventory Events

Search

manager name: wasuh.manager

rule groups: syscheck

agent id: 001

Add filter

DQL

Last 24 hours

Show dates

Refresh

Most active users

root (100%)

Actions

modified (100%)

Events

modified

Files added

No results found

Files modified

/etc/hosts.conf

Files deleted

No results found

W. File Integrity M. kal\_mol

Dashboard Inventory Events

Search

manager name: wasuh.manager

rule groups: syscheck

agent id: 001

Add filter

DQL

Last 24 hours

Show dates

Refresh

Files (7509)

File

Last modified

User

User ID

Group

Group ID

Size

|                                     |                             |      |   |      |   |         |
|-------------------------------------|-----------------------------|------|---|------|---|---------|
| /bin                                | Mar 20, 2026 @ 22:55:51.000 | root | 0 | root | 0 | 7       |
| /boot/system.map-6.12.38-kali-amd64 | Aug 12, 2025 @ 23:03:25.000 | root | 0 | root | 0 | 83      |
| /boot/config-6.12.38-kali-amd64     | Aug 12, 2025 @ 23:03:25.000 | root | 0 | root | 0 | 283402  |
| /boot/grub/fonts/unicode.pf2        | Mar 20, 2026 @ 23:08:21.000 | root | 0 | root | 0 | 2411806 |
| /boot/grub/grub.cfg                 | Mar 20, 2026 @ 23:08:23.000 | root | 0 | root | 0 | 7102    |
| /boot/grub/grubenv                  | Mar 20, 2026 @ 23:08:21.000 | root | 0 | root | 0 | 1024    |
| /boot/grub/386-pc/915resolution.mod | Mar 20, 2026 @ 23:08:21.000 | root | 0 | root | 0 | 7820    |
| /boot/grub/386-pc/acpi.mod          | Mar 20, 2026 @ 23:08:21.000 | root | 0 | root | 0 | 10576   |
| /boot/grub/386-pc/loader32.mod      | Mar 20, 2026 @ 23:08:21.000 | root | 0 | root | 0 | 1200    |
| /boot/grub/386-pc/affs.mod          | Mar 20, 2026 @ 23:08:21.000 | root | 0 | root | 0 | 5644    |
| /boot/grub/386-pc/afs.mod           | Mar 20, 2026 @ 23:08:21.000 | root | 0 | root | 0 | 6048    |
| /boot/grub/386-pc/ahci.mod          | Mar 20, 2026 @ 23:08:21.000 | root | 0 | root | 0 | 1508    |
| /boot/grub/386-pc/ahci.mod          | Mar 20, 2026 @ 23:08:21.000 | root | 0 | root | 0 | 15608   |
| /boot/grub/386-pc/ahci.mod          | Mar 20, 2026 @ 23:08:21.000 | root | 0 | root | 0 | 492     |
| /boot/grub/386-pc/ahci.mod          | Mar 20, 2026 @ 23:08:21.000 | root | 0 | root | 0 | 956     |
| /boot/grub/386-pc/ahci.mod          | Mar 20, 2026 @ 23:08:21.000 | root | 0 | root | 0 | 3024    |
| /boot/grub/386-pc/ahci.mod          | Mar 20, 2026 @ 23:08:21.000 | root | 0 | root | 0 | 4116    |
| /boot/grub/386-pc/ahci.mod          | Mar 20, 2026 @ 23:08:21.000 | root | 0 | root | 0 | 5544    |
| /boot/grub/386-pc/ahci.mod          | Mar 20, 2026 @ 23:08:21.000 | root | 0 | root | 0 | 1576    |
| /boot/grub/386-pc/ahci.mod          | Mar 20, 2026 @ 23:08:21.000 | root | 0 | root | 0 | 6764    |

W. File Integrity M. kal\_mol

Dashboard Inventory Events

Search

manager name: wasuh.manager

rule groups: syscheck

agent id: 001

Add filter

DQL

Last 24 hours

Show dates

Refresh

timestamp per 30 minutes

Count

1

0.8

0.6

0.4

0.2

0

04:00

1

timestamp per 30 minutes

00:00

03:00

06:00

09:00

12:00

15:00

18:00

1 hit

Apr 30, 2026 @ 20:59:22.984 - May 1, 2026 @ 20:59:22.984

Export Formatted

710 available fields

Columns

Density

1 fields sorted

Full screen

| timestamp                  | agent.name | syscheck.path   | syscheck.event | rule.description            | rule.level | rule.id |
|----------------------------|------------|-----------------|----------------|-----------------------------|------------|---------|
| May 1, 2026 @ 04:18:11.293 | kal_mol    | /etc/hosts.conf | modified       | Integrity checksum changed. | 7          | 550     |

Demo 3: FIM Alert: Modification to /etc/hosts Detected with Hash Comparison

## Results Discussion

The FIM module detected a modification to `/etc/hosts` on agent `kali_moi`, performed by the root user. While a single file change may appear minor in isolation, modifications to `/etc/hosts` are a known attacker technique used for DNS hijacking, redirecting legitimate domain names to malicious IP addresses without touching the actual DNS infrastructure. The alert captured the exact timestamp, the responsible user account, and flagged it as an integrity checksum change. In an ethical hacking context, this demonstrates how FIM serves as a critical forensic layer: even if an attacker successfully gains access and makes changes quietly, Wazuh records the evidence. A defender reviewing this alert would immediately investigate whether the change was authorized and what domains were affected.

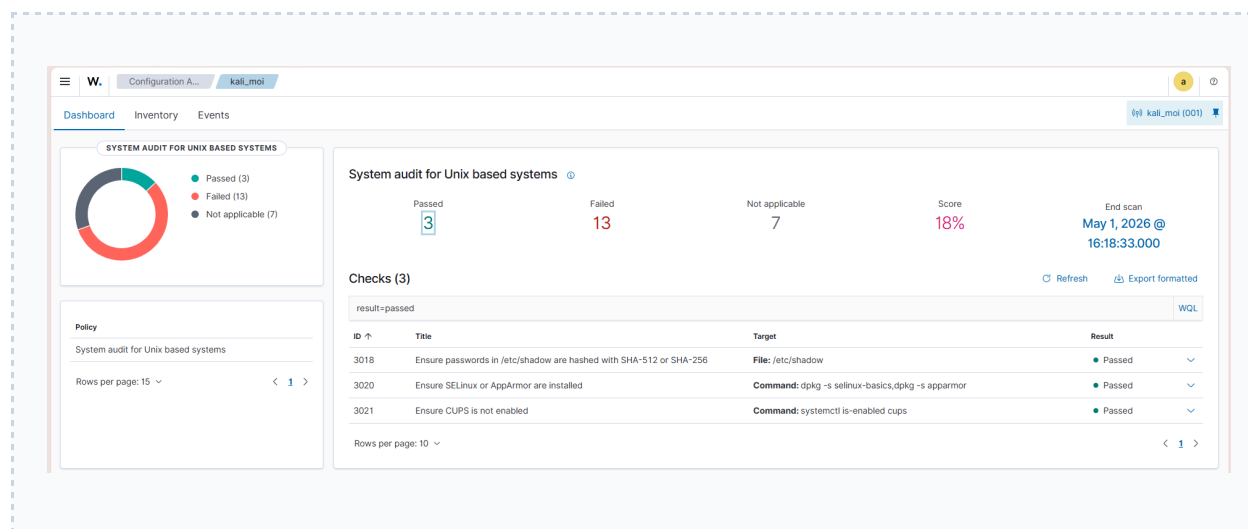
## Demo 4: Security Configuration Assessment (SCA)

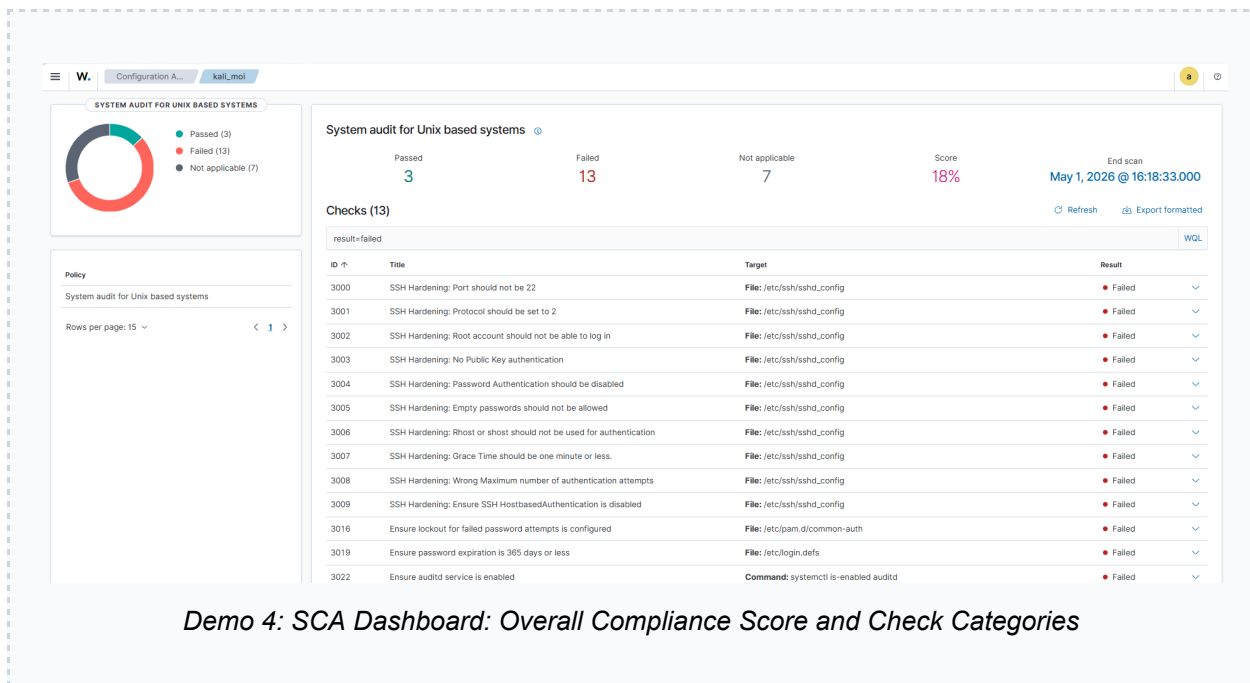
### Objective

Show Wazuh's ability to automatically audit endpoint security configuration against CIS benchmarks and produce an actionable compliance score with specific pass/fail findings.

### Demonstration Steps

1. Navigate to Endpoint Security > Configuration Assessment in the Dashboard.
2. Select an active agent to view its SCA audit results.
3. Review the overall compliance score (percentage of checks passed).
4. Expand individual check categories to see: check description, expected value, actual value, result, and remediation command.





*Demo 4: SCA Dashboard: Overall Compliance Score and Check Categories*

## Results Discussion

The SCA audit of agent `kali_moi` produced an overall compliance score of 18% against the CIS Unix benchmark, with 3 checks passed, 13 failed, and 7 not applicable. The failed checks reveal a deeply hardening-deficient configuration: root login over SSH is permitted, public key authentication is disabled, password authentication is enabled, empty passwords are allowed, and no lockout policy exists for failed login attempts. From an ethical hacking perspective, this SCA report essentially functions as a pre-written attack surface map, every failed check is a potential entry point or privilege escalation vector. The 18% score indicates that this endpoint would be highly susceptible to the techniques demonstrated in Demo 2, directly connecting the configuration weaknesses to the real detections observed.

## Demo 5: MITRE ATT&CK Framework View

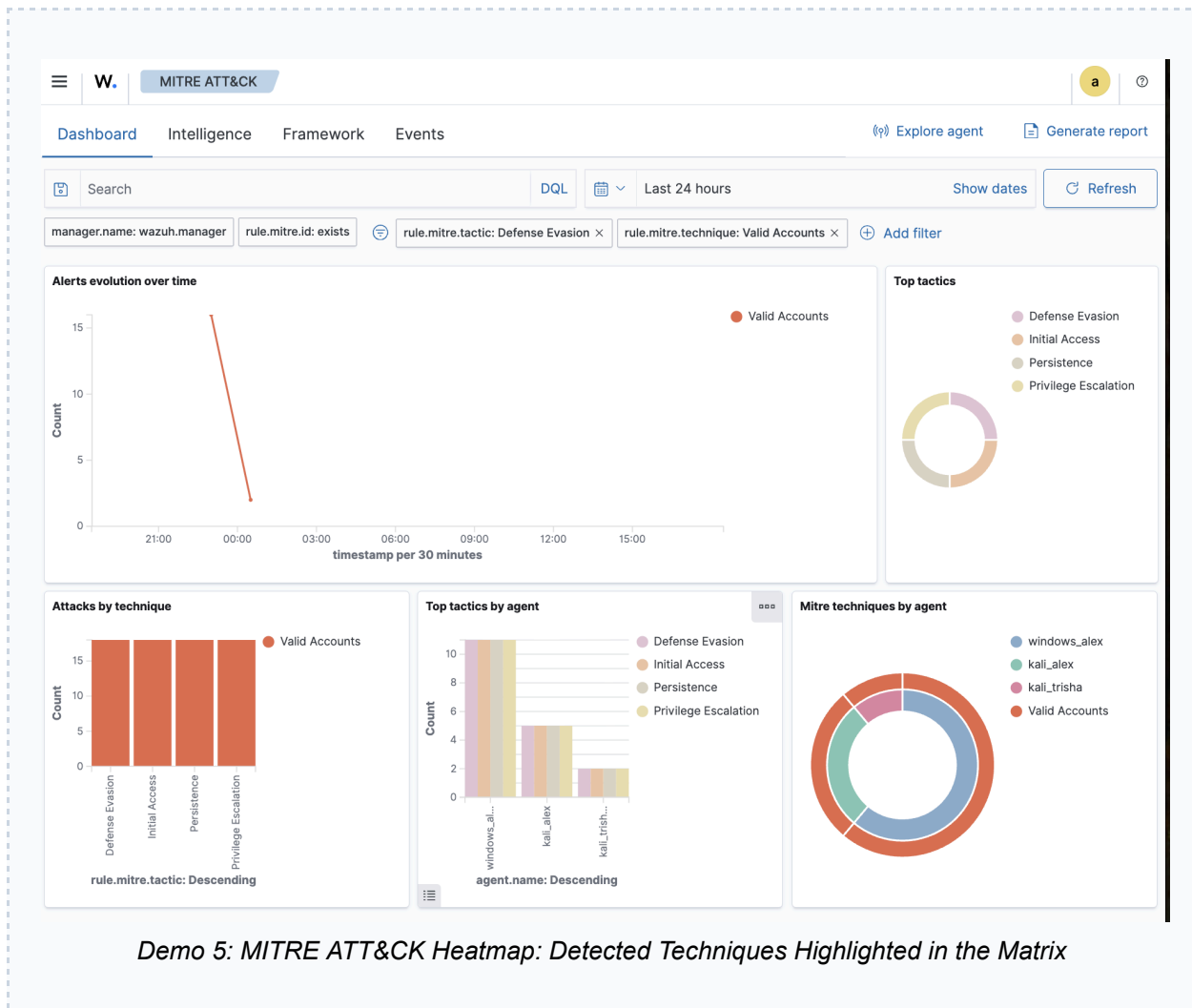
### Objective

Demonstrate how Wazuh contextualizes all detected threats within the MITRE ATT&CK framework, enabling security teams to understand the broader attack campaign and adversary behavior patterns behind individual alerts.

### Demonstration Steps

1. Navigate to Threat Intelligence > MITRE ATT&CK in the Dashboard.
2. View the ATT&CK matrix heatmap showing which tactics and techniques have been detected.
3. Click on a highlighted technique cell to view the list of related Wazuh alerts.

#### 4. Show how the brute force alert from Demo 2 maps to Credential Access > T1110.



## Results Discussion

The Framework tab presented the full ATT&CK matrix with detected techniques highlighted across the kill chain, covering tactics from Initial Access through to Impact. The heatmap view confirmed that activity was distributed across multiple agents and multiple tactic categories, indicating that the monitored environment experienced a broad range of suspicious behaviors rather than isolated incidents. The correlation between agents, kali\_alex generating the highest volume of Defense Evasion events, windows\_alex showing consistent Privilege Escalation activity, and kali\_moi producing the most serious individual findings, paints a picture of a multi-vector threat landscape. For a security operations team, this view would be the starting point for prioritizing incident response: identifying which agents require immediate attention and which tactics represent the most active threats at any given time.

# Advantages & Limitations

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## Advantages

- Completely Free & Open-Source: No licensing fees. Full source code access. Actively maintained by Wazuh Inc. and a global open-source community.
- Unified Platform: Combines SIEM, XDR, EDR, FIM, Vulnerability Detection, SCA, and Compliance monitoring into a single integrated solution, eliminating the need for multiple disparate tools.
- Highly Scalable: Supports single-node Docker deployments (as in this project) up to distributed multi-node clusters handling millions of events per day.
- Broad Agent Support: Native agents available for Linux, Windows, macOS, Solaris, AIX, and HP-UX operating systems.
- Enterprise Integrations: Native integrations with AWS, Azure, GCP, Docker, Kubernetes, GitHub, Slack, PagerDuty, Jira, and more.
- MITRE ATT&CK Mapped: All detections are mapped to the industry-standard threat intelligence framework used by red and blue teams globally.
- Active Community: 10 million+ downloads, active GitHub repository (7,000+ stars), official Slack channel, and paid enterprise support options.
- Regulatory Compliance Coverage: Automated dashboards for PCI DSS, GDPR, HIPAA, and NIST 800-53, dramatically reducing compliance reporting effort.

## Limitations

- Resource Intensive at Scale: The Wazuh Indexer (OpenSearch) requires significant RAM (4-8 GB minimum) and CPU. Scaling to large environments demands careful capacity planning.
- Complex Advanced Configuration: Writing custom decoders, rule sets, and active response scripts requires deep technical knowledge of XML and Wazuh internals.
- Learning Curve: The platform's comprehensive feature set presents a steep learning curve for security analysts who are new to SIEM platforms.
- Docker Networking Complexity: While Docker deployment simplifies setup, it introduces networking complexity for agent-to-manager communication in certain cloud and NAT environments.
- Agent Dependency: Full endpoint visibility requires agent installation on each endpoint. Agentless monitoring options are limited and less capable.
- No Commercial Support by Default: Enterprise support is available but requires a paid subscription. Community support can be slower for complex issues.



## Conclusion

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Wazuh v4.10.1 is a comprehensive, production-grade, and enterprise-quality security platform that delivers remarkable capabilities as a completely free and open-source solution. Through this project, we successfully deployed a fully functional Wazuh environment using Docker on a Debian Linux cloud virtual machine, achieving a single-node stack with all components operational: Manager, Indexer, Dashboard, and connected Agents.

The live environment demonstrates Wazuh's core value proposition: unified security visibility across endpoints, infrastructure, and cloud services from a single, intuitive dashboard. With 4 active agents connected and 74 medium-severity alerts generated in the last 24 hours, the platform is actively monitoring our environment and providing real-time actionable security intelligence.

In the context of IT Elective 3 Ethical Hacking, Wazuh provides invaluable insight into the blue team defensive perspective. Understanding how a SIEM platform detects, logs, and responds to common attack techniques is essential knowledge for ethical hackers and penetration testers. It allows security professionals to design more realistic attack simulations, understand detection coverage gaps, and deliver more actionable findings to their clients.

We recommend Wazuh as a primary security monitoring platform for educational institutions, small-to-medium enterprises, and any security professional seeking enterprise SIEM capabilities without licensing costs. Its rich feature set, active community, and integration with the MITRE ATT&CK framework make it one of the most complete open-source security tools available today.

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